

AZIM RAHAT

347-570-7950 | azimrahat108@gmail.com | [LinkedIn](#) | [GitHub](#) | azimrahat.com

EDUCATION

The City College of New York, The City University of New York

Graduation December 2026

B.S. Computer Science | 3.7 GPA

- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Database Systems, Systems Design, Software Engineering, Computer Architecture, Statistics, Image Processing, Machine Learning

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, Java, SQL

Frameworks: React, Node.js, Express, Next.js, Flask, Django, Tailwind CSS

Libraries: Scikit-learn, Pandas, TensorFlow, Selenium, GraphQL

Databases/Tools: PostgreSQL, MongoDB, MySQL, Redis, AWS, Docker, Git, Kafka

EXPERIENCE

OnePay

June 2026 – Aug 2026

Software Engineer Intern

New York, NY

- Built an internal voice-agent QA platform in place of a **\$60k/yr** vendor tool, streaming live audio over **ElevenLabs ConvAI WebSockets** so an LLM-driven simulated caller runs test scenarios against a customer-support voice agent
- Increased voice-agent eval throughput **10x**, automating manual phone-call testing and post-call grading via an LLM judge
- Engineered simulated caller with configurable audio behavior (accented speech, background noise, interruptions), surfacing production edge cases that drove guardrail and tool-call fixes
- Automated eval test-case creation from production conversations, replaying each interaction's tool calls as saved mocks under existing **PII** controls and cutting creation time from **~5 minutes to seconds**

STEMKasa

May 2024 – May 2025

Software Engineer Intern

New York, NY

- Implemented **RESTful APIs** with **Flask** and **Express** integrating **GPT-4** and **Whisper** to power voice-based Q&A, personalized practice-problem generation, and automated grading for a learning assistant
- Reduced average API response time by **22%** by introducing a **Redis** caching layer in front of frequently-accessed **MongoDB** queries, improving concurrent request handling under load
- Authored **50+** unit and **e2e** tests integrated into a **CI/CD** pipeline, catching regressions and reducing pre-deployment failures
- Shipped **10+** reusable **React** components (forms, dashboards, inputs) adopted team-wide, accelerating frontend delivery

Google

Feb 2025 – May 2025

Software Engineering Fellow

New York, NY

- Competitively selected from 350+ applicants; received 1:1 mentorship from Google engineers on system design and algorithms

PROJECTS

Jumble | *WebSockets, Redis, Docker, AWS, React, TypeScript, Express, PostgreSQL*

[live site](#) | [github](#)

- Engineered a real-time multiplayer platform supporting 8-player lobbies, room codes, and synchronized scoring/timer logic through **WebSockets** and **Redis caching** for responsive state updates
- Deployed scalable backend with **Dockerized** services (backend, PostgreSQL, Redis) and automated database migrations, integrating a GitHub Actions **CI/CD** pipeline for reliable deployments on **AWS (RDS, EC2)**
- Grew to **110+** active users post-launch, validating system reliability under concurrent real-time load

GPU-Accelerated Recommender System | *RAPIDS (cuDF, cuML), CUDA, Python, Dask, Pandas, FastAPI*

[github](#)

- Built a GPU-accelerated recommender pipeline over **571M** Amazon reviews and **600k+** items using **NVIDIA RAPIDS**, achieving **5.15x** end-to-end speedup over CPU baseline
- Engineered a custom memory layout with **NVIDIA RMM** and ID remapping, fitting a 26 GB sparse matrix on a 24 GB GPU and reclaiming 4-6 GB of VRAM
- Trained a matrix factorization model on **31M** user-item interactions with **59x** faster GPU training, served at **1.4ms** median latency via **FastAPI**

AWARDS & EXTRACURRICULARS

- **Dean's List** (Fall 2022 – Spring 2025), **Grove School of Engineering Scholarship** (2023 – 2025)
- **Board Member, ACM (CCNY)** – Organized coding workshops, technical talks, and peer mentoring sessions for CS students